PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Mentoring as an intervention to promote gender equality in academic medicine: a Systematic Review
AUTHORS	House, Allan; Dracup, Naila; Burkinshaw, Paula; Ward, Vicky; Bryant, Louise

VERSION 1 – REVIEW

REVIEWER	Pooja Gangwani University of Rochester, Rochester, New York, USA
REVIEW RETURNED	17-Jun-2020

GENERAL COMMENTS	Very pertinent and important research topic.

REVIEWER	Dr Imogen Coe Ryerson University Canada
REVIEW RETURNED	18-Jun-2020

GENERAL COMMENTS	This is an important and overdue study on the evidence in support of the role and impact of mentoring in advancing gender equality in academic medicine. I often hear about mentoring programs as being effective actions of change within an organization (the deficit model for "fixing" the target group rather than focusing on organisational change). I respond by informing institutions & individuals on the paucity of data in support of such programs. However, human psychology and almost all organizational cultures assume value in such programmes, without evidence, and a thorough academic analytical review is thus welcomed. The methodology is rigorous and appropriate. The findings are clearly presented and well discussed. Having said that, there are some minor revisions that I think will bring a wider readership and added clarity to the contribution. Please make it clearer at the beginning that the analysis took into account intersectionality of mentoring programmes - such that mentoring programs were assessed for being specific to minority ethnic groups or similar (LGBTQ2S+, disabilities, Black etc.) Then report these data if possible - what proportion of the reports actually pay attention to this? This was not clear until later in the paper and initially appeared to be a deficit in the study until I was well into the body of the text. Also - I assume the review is global in scope and some reference to the distribution of sources internationally would be useful. Are UK and US studies over-represented (for obvious linguistic & demographic reasons) - and thus are the results skewed within a cultural context of those two medical cultures of mentoring? Indeed, there was no discussion of international variations in

	mentoring - or is the general vagueness in systems, approaches and outcomes is universal. For instance, hierarchical gendered structures are entrenched in academic medicine in some places - does this correlate with a lack of attention or an increased attention to mentoring as an approach to shifting numbers? Mentoring as a cultural intervention seems to be more prevalent in some parts of the world (reflecting societal attitudes) compared to others but whether this is real or simply perception, I don't know. This review would seem to have the data to ask this question (along with the question of the intersectional - or not - nature of these programs). All to say - the authors have compiled an impressive dataset and is it possible to derive more information from it. Please provide definitions of terms - for instance, the term "staff" in a north american context refers to administrative or technical support staff within an institution. Use of the term "faculty" refers to the equivalent (I think) of the term "staff" in the UK context and as used in this contribution (e.g. Bottom of page 13). In some jurisdictions there are mentoring schemes for staff and for faculty. They are distinct non-overlapping employment categories in some countries and not in others. Adding clarity & precision in use of terms here will ensure understanding by an international audience. There is clearly a gap between perception of efficacy (perhaps self-efficacy) and actual change. Can the authors comment on this? Also, the costs associated with the embracing the deficit model to the extent that mentoring encourages women to act more like men - and the social costs associated with thus challenging gender norms on two axes - as women in medicine and as assertive, confident (or rather, aggressive, self-serving). There is a substantial literature on this and this is a major criticism of mentoring programming generally. Similarly, the power of sponsorship, rather than mentoring - which may lead to gender equality but not as much self-perceived positive impact. All to say - this reviewer would welcome more discussion comparing and contrasting the role of mentoring relative to other interventions. Similarly, I was expecting a more fulsome discussion of mentoring as an organisational strategy for gender equality compared to other approaches. This was briefly touched on in the discussion. Are there any metrics to demonstrate what has actually made a difference - perhaps in sectors other than academic medicine. For instance, there is a large body of literature from the business world on this issue.

REVIEWER	Zoe Jordan JBI, University of Adelaide, Australia
REVIEW RETURNED	03-Sep-2020

GENERAL COMMENTS	First let me commend you on a deserving topic of exploration and the considerable amount of work you have done. I have some comments (below) which I hope will assist you to strengthen your manuscript for publication. My major concern is around the articulation of the systematic review methodology employed and I apologise if I have misconstrued the steps that were followed. You do state that this work follows a systematic review (SR)

	methodology, however you then go on to state that you utilised the TIDieR template. While I am not deeply familiar with this template, my understanding is that while it can be used as a useful adjunct to enhance the SR process, but it is not a SR methodology unto itself (like that prescribed by Cochrane for example). Given that there was no critical appraisal or synthesis I am not sure you can refer to this work as an SR. It would be good to have some transparency around why these decisions were made (i.e. not to appraise the studies for quality). You state that you "noted study design as an indicator of rigour in evaluation research", which was a considerably problematic statement for me, given the issues we see consistently in the quality of reporting. It is also unclear what the relationship between this work and the previous review, undertaken by Sambunjak and colleagues is. Is this an update of that review or a replication? Did it follow the same methodology as the original review (inclusion/exclusion criteria etc)? If it did not, then I am unclear as to why you would only include studies post 2006? From a replication perspective these are important issues to provide clarity around so that readers know if this review purposefully repeats the same methods in order to verify one or more empirical findings OR to purposefully extend or narrow the review to a broader or more focussed question. An update, however, would simply mean the same review but with studies from 2006 onwards included and I don't think this is clearly articulated in the current manuscript (which is a critical determinant of research waste!). Ultimately, in my view, this is a narrative literature review rather than a systematic review. If it is reframed in this light then I think you will have more opportunity for publication. A couple of other minor queries: The exclusion of LOTE studies is increasingly debated. Do you think that, given your knowledge of the literature, there was potential to have missed critical papers or to have skewed results? Also, I understand that all mentees were female, but I didn't get a sense for who the mentors were? Were they also female or largely male? You hint at this in item 5 but it doesn't appear to be explicit (apologies if I missed this). What potential implications does this have, if any, for future programs going forward or was this difficult to determine? I only ask because I would anticipate that the advice and mentorship provided by a male to a female might be different to that of a female mentor (but perhaps this should be a recommendation for future primary research in the field!). Thank you for undertaking this work and I hope the feedback I have provided is constructive and useful.

VERSION 1 – AUTHOR RESPONSE

Introduction

Reviewer	No	Comment	Precise changes made to the text in response to point including the modified text &	Author

			page numbers in the manuscript where the changes appear	
		No comments from any reviewer		

Methods

Reviewer	No.	Comment	Precise changes made to the text in response to point including the modified text & page numbers in the manuscript where the changes appear	Author
2		Please make it clearer at the beginning that the **analysis** took into account intersectionality of mentoring programmes - such that mentoring programs were assessed for being specific to minority ethnic groups or similar (LGBTQ2S+, disabilities, Black etc.) Then report these data if possible - what proportion of the reports actually pay attention to this? This was not clear until later in the paper and initially appeared to be a deficit in the study until I was well into the body of the text.	We found no studies, even those that reported on schemes aimed at improving outcomes for minority ethnic groups, that reported results relevant to questions of intersectionality. We have added text to the results to clarify this point (Lines 501-503). “We found no study that examined the impact of mentoring in those populations defined by what is usually called intersectionality – that is, schemes where the focus might be for example on female staff from minority ethnic backgrounds, or living with a disability.”	
2		Please provide **definitions of terms** - for instance, the term "staff" in a north american context refers to administrative or technical support staff within an institution. Use of the term "faculty" refers to the equivalent (I think) of the term "staff" in the UK context and as used in this contribution (e.g. Bottom of page 13). In some jurisdictions there are mentoring schemes for staff and for faculty. They are distinct non-overlapping employment categories in some countries and not in others. Adding clarity & precision in use of terms here will ensure understanding by an international audience.	We only included studies where it was clear that academic and academic clinical staff received mentoring. We have added text in methods and results to make this point clear. We added the word “academic” in the inclusion criteria: (Lines 226-227) “a mentee (sometimes called a protégé) defined as a more junior/inexperienced colleague who was a member of academic staff” And in results we have	

			added: (Lines 405-407) “In this review we included only those reports where it was clear that the staff referred to (using whatever terminology the authors employed) were in academic rather than managerial or administrative positions.”	
3		You do state that this work follows a systematic review (SR) methodology, however you then go on to state that you utilised the TIDieR template. While I am not deeply familiar with this template, my understanding is that while it can be used as a useful adjunct to enhance the SR process, but it is not a SR methodology unto itself (like that prescribed by Cochrane for example). Given that there was no critical appraisal or synthesis **I am not sure you can refer to this work as an SR.**	Our review meets the criteria for a systematic review as proposed by a recent review of the diverse terminology used in this area. We have changed the format of the methods section to make this point clearer. (Lines 181-182) “There is no standardized definition of what constitutes a systematic review: here we use the six criteria suggested by Martinic and colleagues, as below.”	
3		It would be good to have some transparency around **why these decisions were made (i.e. not to appraise the studies for quality**). You state that you “noted study design as an indicator of rigour in evaluation research”, which was a considerably problematic statement for me, given the issues we see consistently in the quality of reporting.	We have changed the wording to clarify this point. (Lines 263-266) **(i)** “**<u>Study quality</u>** Because of the wide range of study designs in the studies we identified, we were not able to apply a single standard assessment of quality. We therefore noted study design, as did Sambunjak et al, as the single most important indication of likely biases in evaluation research. We did not exclude any studies based solely upon research	

			design, if they met the eligibility criteria noted above." (Lines 278-280) "we found 20 primary observational studies [30-49] which were either cross-sectional or before-and-after studies with no study using interrupted time series or similar analysis, and we identified four RCTs"	
3		It is also unclear what the **relationship between this work and the previous review, undertaken by Sambunjak and colleagues is**. Is this an update of that review or a replication? Did it follow the same methodology as the original review (inclusion/exclusion criteria etc)? If it did not, then I am unclear as to why you would only include studies post 2006? From a replication perspective these are important issues to provide clarity around so that readers know if this review purposefully repeats the same methods in order to verify one or more empirical findings OR to purposefully extend or narrow the review to a broader or more focussed question. An update, however, would simply mean the same review but with studies from 2006 onwards included and I don't think this is clearly articulated in the current manuscript (which is a critical determinant of research waste!).	Since the Sambunjak review showed clearly that there was no methodologically robust evidence for mentoring in this context, our question was – has any new evidence emerged since 2006 to change that conclusion? We have changed the text in the introduction to make this point clearer, and to indicate that we did not simply replicate the previous review's methods.. (Lines 158-165) "Since the exhaustive systematic review of Sambunjak *et al* [17] found no evidence to support the efectiveness of mentoring interventions in reducing gender inequalities, the present systematic review aimed to identify evidence published since that date, with a particular focus on answering the question - is there new evidence (published since 2006) about whether organised mentoring schemes reduce gender inequalities in academic medicine. In undertaking this review we did not simply replicate the methods used by Sambunjak et al, but took the opportunity to ask a subsidiary question about whether it is feasible to consider mentoring as analogous to so-called complex interventions in healthcare."	
3		Ultimately, in my view, this is a **narrative literature review rather than a**	We have changed the wording in the methods'	

		systematic review. If it is reframed in this light then I think you will have more opportunity for publication.	section to make it clearer that our use of the TIDiER framework was to aid data extraction and synthesis, which is compatible with systematic review methods. (Lines 270-272) "Data were extracted to a standard form designed using the TIDieR framework as a template (see Table 1 for a summary of the framework). We then used this framework as the basis for a narrative synthesis of the results from identified studies."	

Results

Reviewer	No.	Comment	Precise changes made to the text in response to point including the modified text & page numbers in the manuscript where the changes appear.	Author
2		I assume the review is global in scope and some reference to the distribution of sources internationally would be useful. **Are UK and US studies over-represented** (for obvious linguistic & demographic reasons) - and thus are the results skewed within a cultural context of those two medical cultures of mentoring?	We have added a section in the discussion, pointing out this limitation and citing Japan as an example of a place that is sufficiently culturally different that our results may not apply, and yet from which we were unable to find an English language evaluation. (Lines 617-624) "It is a limitation of our review that we included only English language publications. To take one example, we know that the problem of gender inequality in medicine is substantial in Japan, where	

			fewer than 10% of medical school professors are women. There, efforts at change appear to have had limited impact but a full evaluation is not available in an English language academic publication. Nonetheless we believe that our findings can be applied to many countries in Europe, to North America and to other countries with similar educational and cultural traditions."	
3		Also, I understand that all mentees were female, but I didn't get a sense for who the mentors were? Were they also female or largely male? You hint at this in item 5 but it doesn't appear to be explicit (apologies if I missed this). What potential implications does this have, if any, for future programs going forward or was this difficult to determine? I only ask because I would anticipate that the advice and mentorship provided by a male to a female might be different to that of a female mentor (but perhaps this should be a recommendation for future primary research in the field!).	We have reviewed our manuscript and added text to the results make clear that most schemes end up with a majority of male mentors because of availability. We had already noted that, contrary to expectation, there is no strong evidence that gender-matching is highly valued compared with matching on other characteristics. (Lines 357-359) "In most schemes, mentors were defined by seniority in the organisation, as senior faculty, professors, experienced researchers, or department heads [55, 60, 63] [54, 67, 71, 72] with several reports noting that this led to a preponderance of male mentors."	

Discussion

Reviewer	No.	Comment	Precise changes made to the text in response to point including the modified text & page numbers in the manuscript where the changes appear	Author
2		Indeed, there was no **discussion of international variations in mentoring** -	We have added text in the discussion to make this	

		or is the general vagueness in systems, approaches and outcomes is universal. For instance, hierarchical gendered structures are entrenched in academic medicine in some places - does this correlate with a lack of attention or an increased attention to mentoring as an approach to shifting numbers? Mentoring as a cultural intervention seems to be more prevalent in some parts of the world (reflecting societal attitudes) compared to others but whether this is real or simply perception, I don't know.	point: (Lines 617-624) "It is a limitation of our review that we included only English language publications. This limits what we can say about the impact of mentoring in countries which may have very different cultures in relation to gender equality, as well as different approaches to medical training and career support .To take one example, we know that the problem of gender inequality in medicine is substantial in Japan, where fewer than 10% of medical school professors are women. There, efforts at change appear to have had limited impact but a full evaluation is not available in an English language academic publication. Nonetheless we believe that our findings can be applied to many countries in Europe, to North America and to other countries with similar educational and cultural traditions."	
2		There is clearly a gap between perception of efficacy (perhaps self-efficacy) and actual change. **Can the authors comment** on this? …	We think it steps beyond the evidence we have reviewed to go further than we already have in commenting on why mentoring remains so often recommended in the face of so little evidence for its efficacy in this respect.	
2		…Also, the costs associated with the embracing the deficit model to the extent that mentoring encourages women to act	We consider that reviewing this literature goes beyond answering the question in	

		more like men - and the social costs associated with thus challenging gender norms on two axes - as women in medicine and as assertive, confident (or rather, aggressive, self-serving). There is a substantial literature on this and this is a major criticism of mentoring programming generally.	our review and would add unreasonably to the length of the manuscript.	
2		Similarly, the power of sponsorship, rather than mentoring - which may lead to gender equality but not as much self-perceived positive impact. - this reviewer would **welcome more discussion comparing and contrasting the role of mentoring relative to other interventions**.	We consider that reviewing this literature goes beyond answering the question in our review and would add unreasonably to the length of the manuscript.	
2		Similarly, I was expecting a **more fulsome discussion of mentoring as an organisational strategy for gender equality compared to other approaches**. This was briefly touched on in the discussion. Are there any metrics to demonstrate what has actually made a difference - perhaps in sectors other than academic medicine. For instance, there is a large body of literature from the business world on this issue.	We consider that reviewing this literature goes beyond answering the question in our review and would add unreasonably to the length of the manuscript.	
3		The exclusion of LOTE studies is increasingly debated. **Do you think that, given your knowledge of the literature, there was potential to have missed critical papers or to have skewed results?**	As noted above, we have added text to the discussion. (Lines 617-624) "It is a limitation of our review that we included only English language publications. This limits what we can say about the impact of mentoring in countries which may have very different cultures in relation to gender equality, as well as different approaches to medical training and career support .To take one example, we know that the problem of gender inequality in medicine is substantial in Japan, where fewer than 10% of medical school professors are women. There, efforts at change appear to have had limited impact but a	

			full evaluation is not available in an English language academic publication. Nonetheless we believe that our findings can be applied to many countries in Europe, to North America and to other countries with similar educational and cultural traditions.”	

VERSION 2 – REVIEW

REVIEWER	Dr Imogen R. Coe Ryerson University, Canada
REVIEW RETURNED	13-Dec-2020

GENERAL COMMENTS	My comments have been sufficiently addressed.